



Hypothetical Investigation for considering Image quality and Data privacy in Medical Imaging

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1. What is the goal of the research?

In this study, we aim to assess the potential impact on medical imaging by employing steganography, a method that conceals information within other data. Our objective is to examine whether the **quality and data privacy** of medical images are affected when **additional information** is embedded using **steganographic** techniques.

2. What is/are the Research Question(s)?

RQ1: Does the quality of pictures is compromised when steganography is used in medical imaging?

RQ2: Does the data privacy is compromised when steganography is used in medical imaging?

3. And, if applicable, what is/are the hypothesis/ses?

Null Hypothesis (H_0): There is no significant impact of steganography on the quality of pictures and data privacy in medical imaging.

Alternative Hypothesis (H_1): Steganography has a significant impact on the quality of pictures and data privacy in medical imaging.

4. What is the research method to be used?

The research method used here is a “**Formal Experiment Method**”.

The "Formal Experiment Method" is used in this study on medical imaging to carefully control and manipulate factors like image quality and data privacy. This helps ensure reliable and objective results by using random assignment, procedures, and standardized conditions. This method allows researchers to draw clear conclusions about how changes in image quality and data privacy impact medical imaging outcomes, providing valuable insights for improving practices in healthcare.

5. Who will participate?

It is a Human-Computer Interaction(HCI) which involves various participants such as:

- HIPAA
- GDPR
- ISO
- NIST
- Digital forensic lab
- Cybersecurity experts

6. What will be measured

(i). Image quality parameters:

S.No	Parameter	Metric
1	Contrast	Contrast-to-Noise Ratio (CNR)
2	Resolution	Mean Squared Error (MSE)
3	Noise	Peak Signal-to-Noise Ratio (PSNR)

(ii). Data Privacy:

S.No	Parameter	Metric
1	Data Hiding	LSB(OpenStego Tool)
2	Compliance with Privacy Regulations and Data Security	Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR), California Consumer Privacy Act (CCPA)

7. How will it be measured?

(I) Image quality parameters will be assessed using the above-mentioned metrics and subjective ratings.

(A) The contrast of the image can be measured using CNR:

$$\text{CNR} = \text{contrast/noise} = \Delta S / N$$

Where, **contrast** is the difference in signal intensity i.e. (ΔS) between two regions of interest(e.g., between a target structure and its surroundings). **Noise** refers to unwanted or random variations in the pixel values of an image.

In medical imaging, such as X-ray, MRI, or CT scans, CNR is often used to evaluate the visibility of structures within the image.

Enhancing image contrast can improve the concealment of medical images, while reducing contrast may reveal the hidden content.

However, a higher Contrast-to-Noise Ratio (CNR) is considered better visibility, as it indicates a stronger signal relative to the background noise.

(B) The resolution of the image can be assessed by measuring the Mean Squared Error:

$$MSE = \frac{1}{N} \sum_{i=1}^N (I_i - K_i)^2$$

N is the total number of pixels in the image.

I_i is the intensity (pixel value) of the i-th pixel in the original image.

K_i is the intensity of the i-th pixel in the reconstructed or compressed image

MSE measures the average squared difference between the pixel values of the original and steganographic images.

Hence a **lower MSE value** is considered **better**.

(C) The noise in an image can be assessed by measuring its PSNR:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right)$$

MAX is the maximum possible pixel value of the image (for example, 255 for an 8-bit image).

MSE is the Mean Squared Error, which is calculated as mentioned earlier.

The PSNR value is expressed in decibels (dB), higher PSNR values indicate better image quality.

Signal: Represents the original image.

Noise: Represents any distortions or errors occurred during compression.

Original Cover Image	Stego Image	Performance Metrics
		MSE = 0.0302 PSNR = 63.3309 dB
		MSE = 0.0063 PSNR = 70.1539 dB
		MSE = 0.0215 PSNR = 64.8082 dB
		MSE = 0.0242 PSNR = 64.2903 dB

Figure 1.Compairison between Original and Stego Image

As evident in this comparison between the original image and the stego image, a lower Mean Squared Error (MSE) and higher Peak Signal-to-Noise Ratio (PSNR) values suggest that the secret message is nearly imperceptible, with minimal observable changes in both images, making them appear virtually identical.

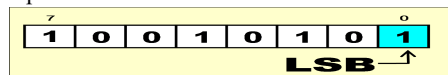
The existence of a secret message is extremely low almost non-existent, due to the low value of the MSE and the high value of the PSNR.

(II) Data Privacy parameters will be assessed using the above-mentioned metrics:

(A) Data Hiding process can be done by Least Significant Bit metric:

$$\text{LSB} = \mathbf{b} [0]$$

where **b [0]** is rightmost bit in the binary representation of the number.



The general value used for LSB replacement in steganography is small to **avoid noticeable changes** in the images.

The secret message is embedded in the medical image's **LSB** via encryption. The medical picture is **first transformed to bytes**, with **each byte's LSB set to zero**. The secret message is then **encrypted** and **transformed** into a **series of bits** that must be placed into the medical image's vacant spaces. Finally, a **stego picture** is produced that has the **same appearance** as the **original medical image** but includes the **secret message** by swapping out the bits of the secret encrypted message in the LSB in each byte of the original medical image.



Figure 2..Medical Report before steganography



Figure 3.Medical Report after steganography

The figures show almost no visible changes, suggesting that the LSB method lets you hide data without making any noticeable differences.

(B)Compliance with Privacy Regulations:

It ensures that steganographic techniques complies with healthcare privacy laws, such as the Health Insurance Portability and Accountability Act (**HIPAA**) in the United States, **GDPR** and **CCPA**.

Medical imaging data is protected by HIPAA, the Health Insurance Portability and Accountability Act, which also guarantees data security and privacy. Following HIPAA regulations is essential when employing steganography in medical imaging. The Security Rule requires data integrity and access controls, the Privacy Rule regulates patient confidentiality, and risk assessments must take steganography into account in order to secure electronic protected health information (ePHI). Data integrity and patient privacy are protected by compliance.

Standards set by organisations like the National Institute of Standards and Technology (**NIST**) provide frameworks for securing information systems, including medical imaging systems.

The use of advanced technologies(such as Artificial Intelligence), automation in implementing security updates and continuous monitoring can benefit medical imaging systems from dynamic security that adapts to threats.

8.What will be the process used for data collection?

The process used for the data collection comprises of the following:

- **Participant Recruitment and Consent:** In this step we recruit different medical imaging experts.
- **Random assignment and blinding:** Randomly assign participants to employ blinding techniques.
- **Introduction and Image Presentation:** Brief Introduction of the study and present the set of medical images, both with and without steganography information.
- **Image Quality and Privacy Assessment:** The most imperative step in the data collection process. Participants are instructed to evaluate the image quality using standardized techniques. In similar fashion, privacy risks are assessed by those participants.
- **Data Record:** Now the quantitative data is recorded from image quality assessments and privacy risk participant's responses.
- **Debriefing:** After the assessments, participants will have debriefing sessions to answer questions that they might have and to clear up anything they find confusing.
- **Data Analysis:** Statistics are considered, such as LSB, CNR, etc. for image quality metrics and correlation analyses for privacy risk assessments.
- **Result Interpretation and Report:** The results are analysed to determine what information can be obtained about the privacy and medical images.

9.How are you planning to analyse the data?

- Initially identifying the patient's information that needs to be protected. Then convert it into a secure and compressed format.
- Choosing appropriate steganography technique which suits medical imaging data such as Least Significant Technique (LSB).
- To enhance data privacy, LSB metric is introduced into practice.
- Integration with medical imaging systems that ensures the compatibility with commonly used formats such as DICOM (Digital Imaging and Communications in Medicine) to maintain interoperability.
- Image Quality and Privacy assessments are conducted by employing distinct metrics for medical images.
- After assessments, validation and testing have been performed.
- Eventually documentation and reporting are done.

10. Any confounding factors? Which ones? How are you mitigating them?

- **Confounding factors:** Distinct participant experience levels in steganography can influence proficiency in concealing or detecting hidden information, highlighting the importance of considering these variations in related activities.

Mitigation: We are going to organize the participants by their experience levels and familiarity with steganographic techniques.

- **Confounding factor:** Data privacy perceptions may be influenced separately by the effectiveness of user authentication measures.

Mitigation: Implement standardized user authentication protocols across all conditions to isolate the impact of steganography on data privacy, independent of variations in authentication methods.

- **Confounding factor:** Inadequate data privacy training increases the risk of unauthorized access and compromises the security of participants' sensitive information. Effective training is crucial for preventing data breaches and ensuring a secure environment.

Mitigation: Conducting regular training sessions to educate participants on the importance of data privacy and proper handling of data. Highlighting the importance of considering these variations in related activities.

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